Jie Zhu

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Research Interests **Agentic AI, VLMs, Reinforcement Learning, VQA, Multimodal**

PUBLICATIONS

Selected Publications

- **FusionAgent: A Multimodal Agent with Dynamic Model Selection for Human Recognition.**
  Jie Zhu, Xiao Guo, Yiyang Su, Anil Jain, and Xiaoming Liu. CVPR26. [Paper] [Project] [☆3]
  Keywords: Agentic AI, MLLMs, Refinforcement Learning
- **Can Textual Reasoning Improve the Performance of MLLMs on Fine-grained Visual Classification?**
  Jie Zhu, Yiyang Su, and Xiaoming Liu. CVPR26 (Findings). [Paper] [Project] [☆8]
  Keywords: MLLMs, Reinforcement Learning, Visual Understanding
- **A Quality-Guided Mixture of Score-fusion Experts Framework for Human Recognition**.
  Jie Zhu, Yiyang Su, Minchul Kim, Anil Jain, and Xiaoming Liu. ICCV25. [Paper] [☆18]
  Keywords: Multimodal, MoE
- **DepthAgent: Towards Better Universal Depth Estimation via Sample-wise Expert Selection.**
  Jie Zhu, Girish Chandar Ganesan, and Xiaoming Liu. NeurIPS 26 submission.
  Keywords: Agentic AI, Depth Estimation, MLLMs

Others

- **ATM: Action Temporality Modeling for Video Question Answering.**
  Junwen Chen, Jie Zhu, and Yu Kong. ACM MM23. [Paper][☆4]
  Keywords: VQA, Action Understanding
- **Fairness-Sensitive Policy-Gradient Reinforcement Learning for Reducing Bias in Robotic Assistance.**
  Jie Zhu, Mengsha Hu, Amy Zhang, and Rui Liu. IEEE ROMAN24. [Paper]
  Keywords: Reinforcement Learning, Fairness
- **Subtoken Image Transformer (SiT) for Generalizable Fine-grained Understanding.**
  Jie Zhu, Minchul Kim, Zhizhong Huang, and Xiaoming Liu. (Under review) [Paper]
  Keywords: Image Tokenization, Fine-grained Understanding
- **SapiensID 2.0: Aligning Human Recognition Foundation Models with Human Perception.**
  Yiyang Su, Jie Zhu, Feng Liu, Anil Jain, and Xiaoming Liu. NeurIPS 26 submission.
  Keywords: VLMs, Human Recognition
- **Discovering Holistic Cues to Detect Human-Centric Forgery**
  Xiao Guo, Jie Zhu, Anil Jain, and Xiaoming Liu. ECCV26 submission. [Paper]
  Keywords: MLLMs, Forgery Detection
- **LocalScore: Local Density-Aware Similarity Scoring for Biometrics**
  Yiyang Su, Minchul Kim, Jie Zhu, et al. TBIOM submission. [Paper]
  Keywords: Face Recognition, Gait Recognition, Person ReID
- **Person Recognition at Altitude and Range: Fusion of Face, Body Shape and Gait.**
  Liu Feng, ..., Jie Zhu, et al. TPAMI submission. [Paper]
  Keywords: Multimodal, Biometrics

EDUCATION

Ph.D. in Computer Science, University of North Carolina at Chapel Hill, Advisor: Prof. Xiaoming Liu. Aug 2026 – Apr 2028
Ph.D. in Computer Science, Michigan State University, 4.0/4.0, Advisor: Prof. Xiaoming Liu. Aug 2023 – Apr 2026
MS in Computer Science, George Washington University, 3.9/4.0, Advisor: Prof. Abdou Youssef. Sep 2021 – May 2023
BS in Computer Science, Northeastern University (China), 3.2/4.0, Advisor: Prof. Yuan Liu. Aug 2016 – Jun 2020

RESEARCH PROJECTS

DepthAgent: Adaptive Expert Selection for Universal Depth Estimation.

- Revealed that **pixel-wise fusion** improves depth estimation in a sample-dependent manner, motivating *adaptive expert selection* across perspective, fisheye, and panoramic inputs.

- Proposed DepthAgent, a GRPO-trained VLM agent that performs sample-wise depth expert selection and fusion via multi-turn tool use, achieving strong performance across diverse camera domains, especially on hard samples.

FusionAgent: A Multimodal Agent with Dynamic Model Selection for Human Recognition.

- Proposed **FusionAgent**, an agentic MLLM that leverages *multi-turn ReAct reasoning* and GRPO for dynamic model selection, introducing an *Anchor-based Confidence Top-k (ACT)* fusion for adaptive multimodal integration.
- **+13.2%** on CCVID, **+7.5%** on MEVID, and **+17.7%** improvement on LTCC benchmark over SoTA, with the same efficiency (2.6s for CoT, 1s for fast answer per sample).

Can Textual Reasoning Improve the Performance of MLLMs on Fine-grained Visual Classification?

- Discovered the performance degradation in CoT under zero-shot evaluation and multiple training paradigms. We uncover a central paradox: the degradation induced by CoT is largely driven by the reasoning length in FGVC.
- **+3.6%** on average accuracy across FGVC benchmarks via the proposed ReFine-RFT with ensemble, accuracy-centric rewards while constraining reasoning, and *MRN* for optimization to enhance the RFT training.

A Quality-Guided Mixture of Score-Fusion Experts Framework for Human Recognition.

- Proposed **QME**, a whole-body biometric recognition framework that leverages a *modality-specific Quality Estimator* with a *pseudo quality loss* for adaptive expert weighting and a *score-triplet loss* for cross-modality score alignment.
- **+5.6%**, **+6.8%**, **+6.2%**, **+12.3%** overall improvement on CCVID, MEVID, LTCC, and BRIAR benchmarks in challenging multi-modal real-world scenarios.

Unified MLLMs for Multimodal Understanding and Generation.

- Proposed a *fine-grained GenEval benchmark* for detailed category (Car models) generation; most Unified MLLMs have a poor result, and "think with image" capability failed to improve performance.
- Implemented BAGEL with GRPO (with FlowGRPO/DanceGRPO) for T2I and MMU tasks.

Subtoken Image Transformer (SiT) for Generalizable Fine-grained Understanding.

- Proposed **SiT** using *Attention-based Token Subdivision (ATS)* for dynamic image tokenization that subdivides discriminative regions into subtokens, and a *two-stage fine-tuning* for adaptive region selection.
- **+2.4%** overall accuracy and **+2.5%** average improvement on novel fine-grained categories gain across Fine-grained benchmarks.

WORK EXPERIENCE

Bosch USA — United States
Research Intern — May 2026 – Aug 2026

- Focus on Autonomous Driving (AD) VQA and action prediction.

Inter-American Development Bank — United States
AI Analytics Consultant — Jun 2023 – Aug 2023

- Engineered multilingual web scraping pipelines (50+ media outlets) using BeautifulSoup and Scrapy.
- Built a **ChatGPT-powered dashboard** for summarization and trend analysis of 10,000+ daily text/video news.

HONORS & AWARDS

Graduate Tuition Fellowship (3 out of 74) — Aug 2022
Third Prize Scholarship (30 out of 302) — Sep 2018 – Jul 2020

SKILLS & SERVICES

Reviewer: TCSVT; TPAMI; FG (2024-2026); IJCBLRR (2024)
Skills: Python, PyTorch, Hugging Face, LaTeX, GitHub, MuJoCo, Unity (AR/VR), HTML
Grants: Wrote grants for UniDisplayAgent (SAMSUNG), object re-identification (IRAPA), and generative waveform (IRAPA) with my advisor.
TAship: Computer Animation (Fall 2022), Computer Graphics II (Spring 2023)
Others: Maintained the CVLab website for 3 years.

SELECTED PATENTS

CN113194348B, "Virtual human lecture video generation method, system, device and storage medium". Jul. 2022.
CN113192161B, "Virtual human image video generation method, system, device and storage medium". Oct. 2022.